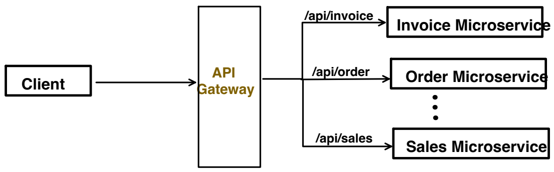
API Gateway

Wednesday, 27 August 2025 10:17 AM

API Gateway:

It provides a Single entry point to access all microservices.



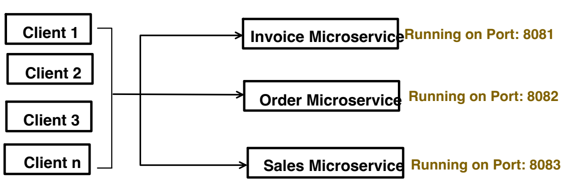
And this Single Entry Point, provides lot of benefits like:

1. **Routing : forward requests to right microservices.**
2. **Load balancing**
3. **Authentication like JWT**
4. **Rate Limiting**
5. **Resilience features (Circuit breaker, Retry etc.)**
6. **Request/Response Transformation**
7. **Monitoring and Logging.**
   Etc..

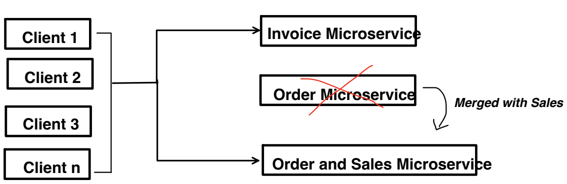
Let's see 1 by 1, how we can achieve all of the above with the API Gateway:

1. Routing : forward requests to right microservices.

Assume this:

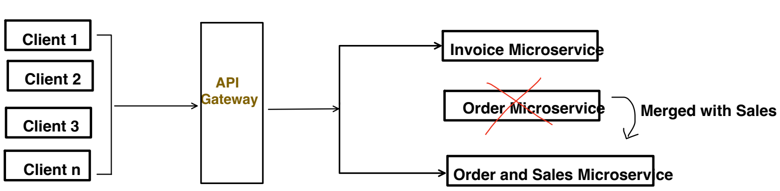


If 1 microservice is further divided into 2 or Multiple microservices merged together, then all clients need to update the routing logic.



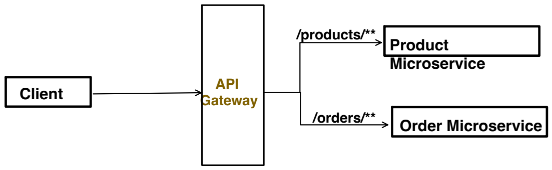
(All clients need to update the routing logic)

But with API Gateway in Mid, changes update required only at 1 service i.e. Gateway



(Only API Gateway need to be updated)

Lets implement this:

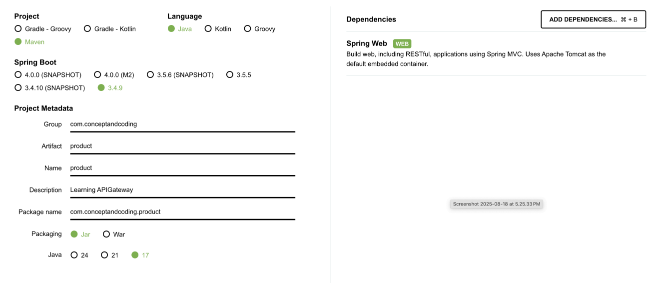


Product Controller Class:

```java
@RestController
@RequestMapping("/products")
public class ProductController {

    @GetMapping("/{id}")
    public ResponseEntity<String> getProduct(@PathVariable String id) {
        return ResponseEntity.ok().body("fetch the product details with id:" + id);
    }
}
```

application.properties

```
server.port=8082
spring.application.name=product-service
```

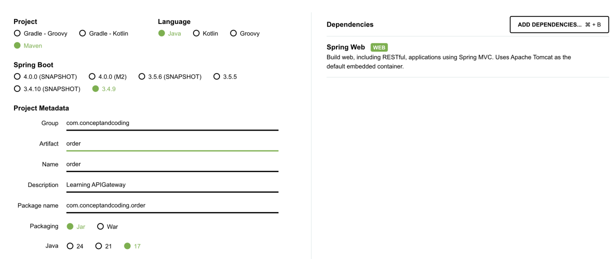


Order Controller Class:

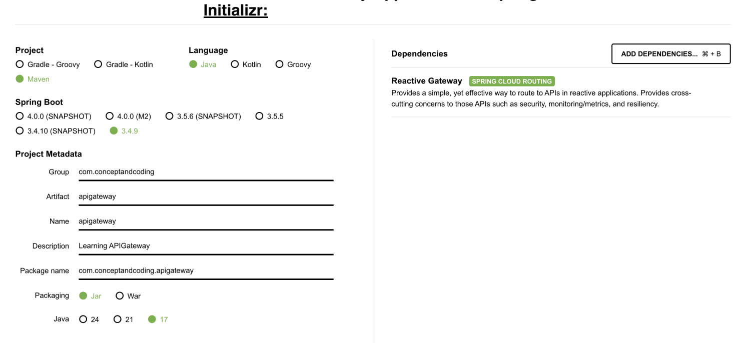
```java
@RestController
@RequestMapping("/orders")
public class OrderController {

    @GetMapping("/{id}")
    public ResponseEntity<String> getOrder(@PathVariable String id) {
        return ResponseEntity.ok().body("fetch the Order details with id:" + id);

    }
}
```

application.properties

```
server.port=8081
spring.application.name=order-service
```



Create a new Gateway Application via Spring Initializr:



In pom.xml, below dependency will get added

```xml
<dependency>
    <groupId>org.springframework.cloud</groupId>
    <artifactId>spring-cloud-starter-gateway</artifactId>
</dependency>
```

application.properties

application.properties

```properties
spring.application.name=apigateway
server.port=8083

# Route: Forward /products/** to product-service
spring.cloud.gateway.routes[0].id=product-service
spring.cloud.gateway.routes[0].uri=http://localhost:8082
spring.cloud.gateway.routes[0].predicates[0]=Path=/products/**

# Route: Forward /orders/** to order-service
spring.cloud.gateway.routes[1].id=order-service
spring.cloud.gateway.routes[1].uri=http://localhost:8081
spring.cloud.gateway.routes[1].predicates[0]=Path=/orders/**
```

Id: This is just a unique name to this route

Uri: Microservice host and port number

predicates: pattern matching to decide the routing.
apart from Path, we can also have predicate on Method like:
spring.cloud.gateway.routes[0].predicates[1]=Method=GET,POST

means GET and POST request to /products/** will match this route.

Start all 3 servers: Product, Order and API Gateway

Product service is running on 8082

```
   main] c.c.p.ProductserviceApplication          : Starting ProductserviceApplication using Java 17.0.
   main] c.c.p.ProductserviceApplication          : No active profile set, falling back to 1 default pr
   main] o.s.b.w.embedded.tomcat.TomcatWebServer  : Tomcat initialized with port 8082 (http)
   main] o.apache.catalina.core.StandardService   : Starting service [Tomcat]
   main] o.apache.catalina.core.StandardEngine    : Starting Servlet engine: [Apache Tomcat/10.1.20]
   main] o.a.c.c.C.[Tomcat].[localhost].[/]       : Initializing Spring embedded WebApplicationContext
   main] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization complete
   main] o.s.b.w.embedded.tomcat.TomcatWebServer  : Tomcat started on port 8082 (http) with context pat
   main] c.c.p.ProductserviceApplication          : Started ProductserviceApplication in 0.583 seconds
-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/]       : Initializing Spring DispatcherServlet 'dispatcherSe
-exec-1] o.s.web.servlet.DispatcherServlet        : Initializing Servlet 'dispatcherServlet'
-exec-1] o.s.web.servlet.DispatcherServlet        : Completed initialization in 0 ms
```

Order service is running on 8081

Run OrderserviceApplication

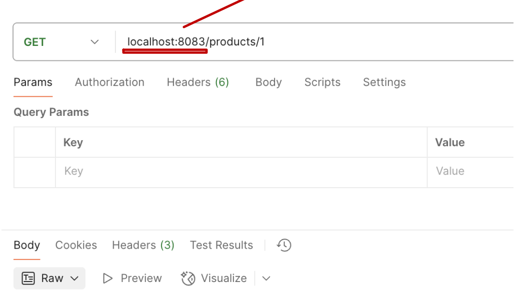
```
der-service] [           main] c.c.o.OrderserviceApplication            : Starting OrderserviceApplication using Java 17.0.
der-service] [           main] c.c.o.OrderserviceApplication            : No active profile set, falling back to 1 default
der-service] [           main] o.s.b.w.embedded.tomcat.TomcatWebServer  : Tomcat initialized with port 8081 (http)
der-service] [           main] o.apache.catalina.core.StandardService   : Starting service [Tomcat]
der-service] [           main] o.apache.catalina.core.StandardEngine    : Starting Servlet engine: [Apache Tomcat/10.1.20]
der-service] [           main] o.a.c.c.C.[Tomcat].[localhost].[/]       : Initializing Spring embedded WebApplicationContex
der-service] [           main] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization comple
der-service] [           main] o.s.b.w.embedded.tomcat.TomcatWebServer  : Tomcat started on port 8081 (http) with context p
der-service] [           main] c.c.o.OrderserviceApplication            : Started OrderserviceApplication in 0.593 seconds
der-service] [nio-8081-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/]       : Initializing Spring DispatcherServlet 'dispatcher
der-service] [nio-8081-exec-1] o.s.web.servlet.DispatcherServlet        : Initializing Servlet 'dispatcherServlet'
der-service] [nio-8081-exec-1] o.s.web.servlet.DispatcherServlet        : Completed initialization in 1 ms
```

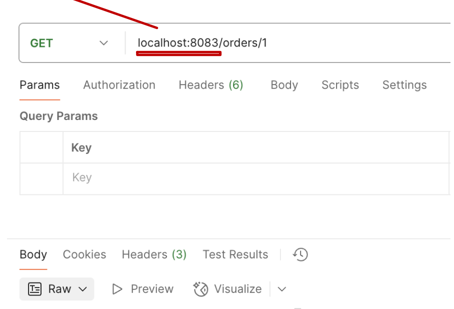
Run ApigatewayApplication

```
[           main] o.s.c.g.r.RouteDefinitionRouteLocator    : Loaded RoutePredicateFactory [Query]
[           main] o.s.c.g.r.RouteDefinitionRouteLocator    : Loaded RoutePredicateFactory [ReadBody]
[           main] o.s.c.g.r.RouteDefinitionRouteLocator    : Loaded RoutePredicateFactory [RemoteAddr]
[           main] o.s.c.g.r.RouteDefinitionRouteLocator    : Loaded RoutePredicateFactory [XForwardedRemoteA
[           main] o.s.c.g.r.RouteDefinitionRouteLocator    : Loaded RoutePredicateFactory [Weight]
[           main] o.s.c.g.r.RouteDefinitionRouteLocator    : Loaded RoutePredicateFactory [CloudFoundryRoute
[           main] o.s.b.web.embedded.netty.NettyWebServer  : Netty started on port 8083 (http)
[           main] c.c.apigateway.ApigatewayApplication     : Started ApigatewayApplication in 0.761 seconds
[ctor-http-nio-3] i.n.r.d.DnsServerAddressStreamProviders  : Unable to load io.netty.resolver.dns.macos.MacO
```

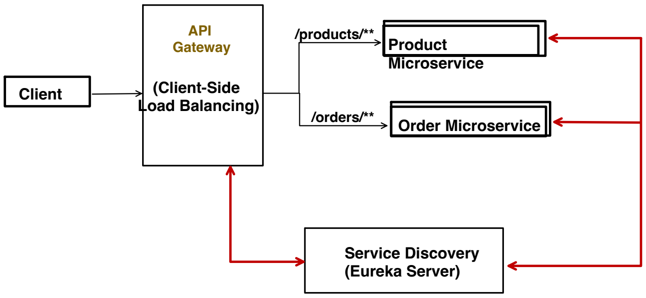
API Gateway service is running on 8083

Invoked API Gateway, and based on PATH matching it invoked the respective microservice.

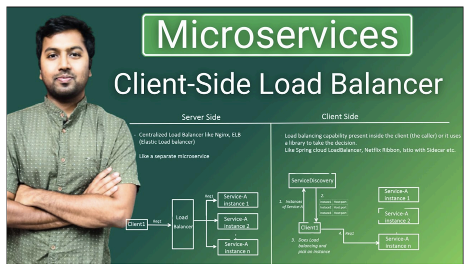




2. API Gateway and Load Balancer



*Already covered **Service Discovery** and **Client Side Load Balancer** topic in this playlist, if you have any doubt with below topics, kindly check it out first.*



Product Service:

Pom.xml

```
<dependency>
    <groupId>org.springframework.cloud</groupId>
    <artifactId>spring-cloud-starter-netflix-eureka-client</artifactId>
</dependency>
```

application.properties

```
application.properties
server.port=8082
spring.application.name=product-service
#path of the Eureka Server
eureka.client.service-url.defaultZone=http://localhost:8761/eureka
```

Controller class

```
@RestController
@RequestMapping("/products")
public class ProductController {

    @GetMapping("/{id}")
    public ResponseEntity<String> getProduct(@PathVariable String id) {
        return ResponseEntity.ok().body("fetch the product details with id:" + id);
    }
}
```

Order Service:

Pom.xml

```
<dependency>
    <groupId>org.springframework.cloud</groupId>
    <artifactId>spring-cloud-starter-netflix-eureka-client</artifactId>
</dependency>
```

application.properties

```
application.properties
server.port=8081
spring.application.name=order-service
eureka.client.service-url.defaultZone=http://localhost:8761/eureka
```

Controller class

```
@RestController
@RequestMapping("/orders")
public class OrderController {

    @GetMapping("/{id}")
    public ResponseEntity<String> getOrder(@PathVariable String id) {
        return ResponseEntity.ok().body("fetch the Order details with id:" + id);

    }
}
```

Eureka Server (Service Discovery):

Pom.xml

```xml
<dependency>
    <groupId>org.springframework.cloud</groupId>
    <artifactId>spring-cloud-starter-netflix-eureka-server</artifactId>
</dependency>
```

application.properties

```properties
application.properties

spring.application.name=eureka-server
server.port=8761

# Since it's a server, we don't want it to register and
# also don't want to fetch the instances details
eureka.client.register-with-eureka=false
eureka.client.fetch-registry=false
```

```java
@SpringBootApplication
@EnableEurekaServer
public class EurekaServerApplication {

    public static void main(String[] args) {

        SpringApplication.run(EurekaServerApplication.class, args);
    }
}
```

API Gateway

Pom.xml

```xml
<dependency>
    <groupId>org.springframework.cloud</groupId>
    <artifactId>spring-cloud-starter-gateway</artifactId>
</dependency>
<dependency>
    <groupId>org.springframework.cloud</groupId>
    <artifactId>spring-cloud-starter-netflix-eureka-client</artifactId>
</dependency>
```

application.properties

```properties
application.properties

spring.application.name=apigateway
server.port=8083

# Route: Forward /products/** to product-service
spring.cloud.gateway.routes[0].id=product-service
spring.cloud.gateway.routes[0].uri=lb://product-service
spring.cloud.gateway.routes[0].predicates[0]=Path=/products/**

# Route: Forward /orders/** to order-service
spring.cloud.gateway.routes[1].id=order-service
spring.cloud.gateway.routes[1].uri=lb://order-service
spring.cloud.gateway.routes[1].predicates[0]=Path=/orders/**


#path of the Eureka Server to fetch the microservices host details
eureka.client.service-url.defaultZone=http://localhost:8761/eureka
```

Replaced the hardcoded URI:
http://localhost:8082

Now internally it will use Spring Cloud LoadBalancer framework to perform the loa balancing.
- first will fetch the instances from Service Discovery and
-Perform Load balancing algorithm